

Binary Stroke Prediction

Stage 2: Pipeline Engineering, Champion Selection & Ablation

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 Colab Notebook: [group11_lab3_stage2.ipynb](#)

 Google Slides: [group11_lab3_stage2](#)

Executive Summary

Stage 2 addresses three Stage 1 gaps: absence of leak-proof sklearn/imblearn Pipelines, multiple test-set evaluations violating holdout discipline, and absence of hyperparameter tuning or ensemble methods. Stage 2 formalises modelling through CRISP-DM, using a single locked test set, all preprocessing encapsulated within pipelines, and up to four controlled ablation experiments. Five models were evaluated via 5-fold stratified CV: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and a Stacking Ensemble.

Stage 1 Baseline Results

Model	Train Recall	Test Recall	Gap	Test AUC
Dummy Classifier	—	0.24	—	0.50
Logistic Regression	0.70	0.56	0.14	0.81
Decision Tree	1.00	0.24	0.76	0.57

The Decision Tree grew to depth 31 with 592 leaves, memorising the training set — a canonical high-variance pathology. Logistic Regression showed moderate bias (test recall 0.56, below the 0.60 target) with counterintuitive coefficients (e.g., `heart_disease` = -0.79), reflecting a linear boundary unable to capture non-linear feature interactions. Precision was ~17% (LR) and ~11% (DT), indicating class imbalance not fully resolved. No hyperparameter tuning or ensemble methods were explored.

Summary of Key Improvements

Stage 2 strengthens the full modelling workflow through three core upgrades: pipeline encapsulation to prevent data leakage, proper holdout discipline with a single locked test set, and systematic hyperparameter tuning plus ablation experiments to isolate the value of each design choice.

Stage 2 Outcomes at a Glance

Champion

Logistic Regression (Stage 1) — CV Recall 0.7037 ± 0.0473 , ROC-AUC 0.8043. Ensembles underperformed due to probability calibration issues on severe class imbalance.

Final Test Recall

0.92 at threshold 0.50; 0.82 at adjusted threshold 0.59 — a 64% improvement over Stage 1's 0.56.

Test ROC-AUC

Test ROC-AUC of 0.8255 — threshold-independent discriminative gain over Stage 1's 0.81.

Overview

- Data engineering details and feature validation
- Pipeline architecture and leakage mitigation
- Model selection via cross-validation
- Ablation experiments and final tuning results

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Data Engineering

Upgrades From The Baseline

All preprocessing is now encapsulated within sklearn/imblearn Pipelines. No statistic is fit on test data. SMOTENC is placed inside the pipeline so synthetic samples are generated exclusively from each CV training fold. Systematic feature validation rules ensure clinical plausibility, logical consistency, and reproducibility across all 10 predictors.

Step	Stage 1 Approach	Stage 2 Upgrade	Rationale
BMI Imputation	Median + missing flag (manual, pre-split)	SimpleImputer(strategy='median') + MissingIndicator inside ColumnTransformer	Fit only on training fold — prevents leakage
OHE (nominal)	pd.get_dummies (manual, pre-split)	OneHotEncoder(drop='if_binary', handle_unknown='ignore')	Handles unseen categories safely; no leakage
Ordinal (smoking)	Manual integer mapping	OrdinalEncoder(categories=['never','formerly','smokes'], encoded_missing_value=-1)	'Unknown' → -1; risk-monotone ordering 0/1/2 preserved
Feature Scaling	StandardScaler applied manually	StandardScaler() inside shared preprocessor	All models receive identical feature representation
SMOTE-NC	Applied after manual encoding, pre-CV	imblearn.pipeline.Pipeline with SMOTENC after preprocessor	Prevents synthetic minority samples from contaminating validation folds
Outlier Removal	Done before split	Retained as-is (age < 1, gender == 'Other')	One-time filter; no transform involves test data

Feature Validation

Clinical standards applied to all 10 predictors — age (1–120 yr), glucose (50–400 mg/dL), BMI (10–60 kg/m²), binary flags [0,1], categorical membership, and cross-feature consistency (e.g., work_type='children' implies age < 18). 16 suspicious BMI values flagged; all other checks passed.

Pipeline Architecture: Mitigating Data Leakage

Critical Engineering Upgrade

The critical engineering upgrade over Stage 1 is full encapsulation: the preprocessor, SMOTENC resampler, and classifier are bundled as a single estimator object. During 5-fold stratified CV, imputation statistics, encoder vocabularies, and SMOTE synthesis are all fit exclusively on each training fold. The test fold and held-out test set are never exposed to any preprocessing fit operation. This is the formal, leak-proof pipeline architecture required by the assignment.

Pipeline Structure



ColumnTransformer Components

- Numerical: SimpleImputer(median) → StandardScaler on age, avg_glucose_level, bmi
- BMI Flag: MissingIndicator → binary bmi_missing feature
- Nominal: OneHotEncoder(drop='if_binary', handle_unknown='ignore') on gender, ever_married, Residence_type, work_type
- Ordinal: FunctionTransformer ('Unknown' → NaN) → OrdinalEncoder (never=0, formerly=1, smokes=2, Unknown=-1)
- Passthrough: hypertension, heart_disease (already binary integers)

SMOTENC Placement Rationale

Placing SMOTENC *inside* the pipeline — not before CV — is the critical leakage fix. In Stage 1, SMOTE was applied globally before cross-validation, contaminating every validation fold's class distribution with synthetic samples derived from the full training set.

In Stage 2, each CV fold generates its own synthetic minority samples from its own training partition only. The held-out test set (1,014 samples, 4.93% prevalence) remains completely isolated until the single final evaluation in Section 6.

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Model Selection

Cross-Validation comparison of 2-3 algorithmic families

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Linear Family

Logistic Regression

Motivation: Stage 1 best performer (AUC 0.81, low variance). Retained as the reference point for comparison.

Expected Behavior: Linear decision boundary provides interpretable coefficients and stable generalization. May underfit complex feature interactions.

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Bagging Ensemble

Random Forest

Motivation: Targets the DT's high variance. Bootstrapped subsets with random feature subsampling reduce sensitivity to individual training observations.

Expected Behavior: Variance reduction through diversity rather than capacity restriction. Individual trees remain deep and expressive while ensemble prediction is stable.

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Boosting Ensemble

HistGradientBoostingClassifier

Motivation: Targets the LR's moderate bias. Sequential residual fitting progressively approximates non-linear boundaries the LR cannot learn.

Why HistGradientBoostingClassifier? Selected over AdaBoost and GradientBoostingClassifier due to: (1) native missing value handling, (2) fast histogram binning, (3) robustness to noise.

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Stacking Ensemble

RF + GBT → LR meta-learner

Motivation: Combines complementary strengths of bagging and boosting. Meta-learner learns which base estimator is more reliable for each region of feature space.

Design: RF and GBT are natural stacking partners — their error profiles are complementary. Logistic Regression as meta-learner is deliberately simple to avoid re-introducing overfitting.

Model	CV Recall (mean)	CV Recall (± std)	CV ROC-AUC (mean)	CV ROC-AUC (± std)	CV Precision	CV F1
Logistic Regression (Stage 1)	0.7037	± 0.0473	0.8043	± 0.0168	0.1335	0.2243
Gradient Boosting	0.2817	± 0.0672	0.7777	± 0.0231	0.1502	0.1943
Decision Tree (Stage 1)	0.2462	± 0.0284	0.5742	± 0.0142	0.1155	0.1570
Random Forest	0.1913	± 0.0575	0.7749	± 0.0157	0.1282	0.1523

Justification for Logistic Regression Model

Key Insight: Linear Signal Dominance

The dataset's signal appears **largely linear**, consistent with LR's strong AUC of 0.80. Despite the theoretical capacity of ensembles to capture complex non-linear interactions, the empirical evidence suggests that the predictive relationships in this stroke dataset are adequately captured by a linear decision boundary.

Random Forest Underperformance Analysis

Unexpected Gap Behavior:

RF's train-test recall gap of **0.81** actually **exceeds** the Decision Tree's gap of 0.75 — the opposite of the expected variance reduction from bagging.

Root Cause: Probability Calibration

RF averages votes across trees, each trained on a 1:1 SMOTE-balanced fold, but outputs **uncalibrated probabilities** at test time on the original ~5% prevalence distribution.

Threshold Misalignment

Default threshold of 0.5 sits in the wrong region of the probability distribution, yielding low recall despite competitive AUC of 0.77.

Conclusion: This is a **threshold calibration problem**, not a failure of the bagging mechanism itself.

Gradient Boosting Limitations

Moderate Recall:

GBT achieved only **0.28 CV recall**, significantly below LR's 0.70. The sequential residual fitting did not translate to improved minority class detection.

Possible Explanations:

1. Default hyperparameters not optimized for severe imbalance
2. Histogram binning may lose fine-grained information critical for rare class discrimination
3. Early stopping not applied, potentially leading to overfitting on SMOTE-synthesized samples.

Stacking Ensemble Failure

Stacking achieved **0.00 recall**, suggesting the meta-learner could not effectively combine the weak base estimators. The complementary error profiles did not materialize in practice.

Key Takeaway: **Complexity does not guarantee performance.** The simplest model (LR) best matched the dataset's underlying structure.

Methodological Note: These findings underscore the importance of empirical validation over theoretical expectations. While ensemble methods offer compelling theoretical advantages for bias-variance trade-offs, their effectiveness is contingent on proper calibration and hyperparameter tuning — investments that were beyond the scope of this Stage 2 comparison.

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Controlled Ablation & Tuning Log

Exp	Description	Change	CV Recall	CV AUC	Delta Recall	Decision
0	LR (Stage 1) — default params	None (baseline anchor)	0.7037	0.8043	0.0000	Baseline
1	LR + RandomOverSampler	Replaced SMOTENC with ROS	0.8088	0.8535	+0.1051	Keep
2	LR + age × hypertension	Added interaction term	0.7035	0.8164	− 0.0002	Reject
3	LR + glucose × BMI	Added interaction term	0.7037	0.8104	0.0000	Reject
4	LR + ROS ratio=1.00	Swept sampling_strategy	0.8088	0.8535	+0.1051	Keep
5	LR + ROS + tuned C/penalty	RandomizedSearchCV	0.9700	0.8365	+0.2663	Keep

Maximum 4 experiments (plus baseline Exp 0). Each changes exactly one variable. All evaluated via 5-fold stratified CV. Test set remains locked throughout.

Cumulative Improvement

Total recall improvement from baseline to final model: **+0.2663** (from 0.7037 to 0.9700), representing a **37.8% relative improvement** .

Key Insights

- Resampling technique had largest single impact (+10.5%)
- Feature engineering interactions provided no benefit
- Hyperparameter tuning provided second-largest gain (+16.1%).

Experimental Discipline

Each experiment changed exactly one variable, enabling clear attribution of effects. The controlled ablation framework ensures reproducibility and scientific rigor.

Final Model Stability Metrics

Per-fold performance of the fully tuned champion pipeline (LR + RandomOverSampler, $C \approx 0.00134$, L1) across all 5 CV folds. Low standard deviation in recall indicates consistent model behaviour — no wide fold-to-fold swings.

0.9700

CV Recall (mean)

± 0.0245 std. across 5 folds

0.8355

CV ROC-AUC (mean)

± 0.0245 std. across 5 folds

0.7414

Balanced Accuracy

± 0.0226 std. across 5 folds

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Non-Zero Coefficients

Out of 150, sparsity rate: 93.3%

Interpretation

Recall std of ± 0.0245 is low — the model is stable across folds. Fold 3 achieved perfect recall (1.0000); Fold 2 was the weakest at 0.9250, still well above the 0.80 clinical floor. ROC-AUC std of ± 0.0192 confirms consistent discriminative ability.

The per-fold consistency validates that the 0.9700 mean recall is not driven by a single favourable fold split.

LI Sparsification Note

Under L1 regularisation at $C \approx 0.00134$ ($\lambda \approx 746$), 14 of 15 post-encoding features are penalised to exactly zero. Only **age** survives with coefficient + 0.6515. This extreme sparsity is the direct consequence of the tuning process selecting very strong regularisation.

Clinically established risk factors such as hypertension, glucose, heart disease, BMI, smoking are assigned zero weight despite known independent contributions to stroke risk. The model is, in effect, an age-thresholding classifier. The ROC-AUC of 0.8255 on the test set confirms this single-feature representation retains substantial discriminative ability which is a testament to how dominant age is as a stroke predictor in this dataset.

Mechanical Failure Analysis

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Row-Level Error Inspection

Extraction Threshold

pred_prob < 0.30 for high-confidence FNs; pred_prob ≥ 0.70 for high-confidence FPs. CV Fold 0 used (not test set) to avoid optimistic in-sample counts. The tuned champion was re-fitted on the fold’s training portion and evaluated on its validation portion.

Confusion Matrix — CV Fold 0 (threshold= 0.50)

True \ Predicted	Pred: No Stroke	Pred: Stroke
True: No Stroke	368 (TN)	403 (FP)
True: Stroke	1 (FN)	39 (TP)

Total FNs: 1 · High-Confidence FNs (prob < 0.30): 0

FN probability range: 0.319 ~ 0.319. The model is not confidently missing strokes.

Total FPs: 403

FP rate: 52.3% of all no-stroke patients flagged.

High-Confidence FPs (prob ≥ 0.70): 63 cases.

Root Cause

This is not a threshold calibration problem — it is a **model capacity problem**. No threshold adjustment can differentiate two patients of the same age; they produce the same raw probability upstream of any threshold logic.

SHAP Local Explanations

False Negative (pred_prob= 0.526, actual= Stroke): The sole contribution comes from age SHAP value (+0.06). If this patient is relatively young, the age contribution is insufficient to cross the model’s implicit threshold.

Had hypertension, glucose, or heart disease survived L1, comorbidity profiles could have contributed additional positive signal. The 7 test-set FNs are precisely stroke cases where age is insufficiently elevated.

High-Confidence FP (pred_prob= 0.744, actual= No Stroke): Driven almost entirely by extreme age. A patient aged 82 is scaled to ~+1.77 standard deviations above the mean; multiplied by age coefficient 0.65, this contributes ~+1.15 to log-odds. Combined with the intercept (shifted positive by 1:1 oversampling), the result is pred_prob ≥ 0.70 regardless of comorbidity profile. Given 5% population prevalence, a raw probability of 0.70 corresponds to a true posterior of only ~10.9%. **Nine out of ten patients flagged at high confidence do not have a stroke.**

High-Confidence FP Pattern

All 8 patients aged 82 receive *identical* predicted probability of 0.731. Both patients aged 81 receive 0.726. These probabilities do not vary despite patients being clinically distinct, namely Row 337 (glucose=215.60, diabetic range) and Row 4490 (glucose=57.56, sub-normal) receive the same score; Row 3167 (hypertension=1) and Row 2325 (hypertension=0) receive the same score. The L1 regularisation has zeroed every coefficient except age, so **predicted probability is a pure function of age** : $p = \sigma(w_{age} \times age_{scaled} + intercept)$. Two patients of the same age receive identical predictions regardless of glucose, blood pressure, cardiac history, or smoking behaviour.

Proposed Fixes for Observed Failures.

1 Reduce Regularisation Strength

Root Cause: 1 at $C \approx 0.00134$ ($\lambda \approx 746$) penalises hypertension, glucose, and heart disease coefficients to zero because their marginal loss reduction does not exceed the penalty.

Targeted fix: Increase C — sweep $C \in [0.01, 0.1, 1.0]$ under the same 5-fold CV protocol. At less aggressive C , these features recover non-zero coefficients, enabling the model to assign meaningfully higher probability to a diabetic 82-year-old (glucose=215.6) than a metabolically healthy one (glucose=57.6).

This would reduce FPs among elderly patients with benign profiles while potentially improving recall for younger stroke patients whose risk is driven by comorbidities rather than age alone — directly addressing both the FP excess and the structural gap producing the single FN.

2 Switch to Elastic Net

L1 alone forces hard sparsity (coefficients exactly zero). Elastic net ($l1_ratio < 1.0$) would allow hypertension, glucose, and heart disease to retain small but non-zero coefficients, recovering clinical signal currently discarded while maintaining regularisation stability. This is a more gradual path than fully removing regularisation.

3 Non-linear Model Class

A gradient boosting model would capture compound interaction effects between age, glucose, and hypertension without requiring explicit interaction engineering or aggressive regularisation. Stroke risk is known to have non-linear, super-additive relationships (e.g., hypertension compounds with age; metabolic syndrome of high BMI + elevated glucose carries disproportionate risk).

The ablation experiments tested age×hypertension and glucose×bmi interaction terms, but neither survived to the final model — regularisation was so aggressive that even genuine interaction signal was penalised away. A non-linear model class would capture these interactions natively.

4 Probability Calibration

Raw predicted probabilities are poorly calibrated by construction: the model was trained on 1:1 oversampled data but evaluated on natural-prevalence (5%) data. Isotonic regression or Platt scaling would align output probabilities with empirical stroke rates, enabling principled threshold selection based on explicit clinical cost functions rather than empirical CV sweeps.

A calibrated probability of 0.70 should correspond to an actual ~70% stroke risk, not ~11%.

Error Cost Analysis

False Negative (FN)

Description: Stroke patient discharged as healthy

Consequence: Delayed treatment → permanent disability or death

False Positive (FP)

Description: Healthy patient flagged as stroke

Consequence: Unnecessary follow-up imaging, patient anxiety, resource use

Clinical Cost Asymmetry

FN >> FP

A missed stroke is categorically more damaging than a false alarm. Treatment within first hours is critical.

Threshold Optimization

Default threshold (0.50) assumes equal FP/FN cost. Given asymmetry, threshold should be lowered to increase recall.

Selected Threshold: 0.59

Fr-maximizing threshold at recall ≥ 0.80 constraint

Final Test Set Evaluation

Metric	Default (0.50)	Adjusted (0.59)
Recall	0.92	0.82
Precision	0.09	0.12
F1-Score	0.16	0.21
Balanced Accuracy	0.71	0.75
ROC-AUC	0.83	0.83

High-Risk Ward

Recommended: Default threshold (0.50)

Prioritize catching all strokes; accept high FP rate for maximum sensitivity

General Screening

Recommended: Adjusted threshold (0.60)

Balance resource constraints with 80% stroke detection rate

Stage 2 vs Stage 1 Improvement

Recall

0.56 → 0.82

+26%

Precision

0.17 → 0.12

-5%

ROC-AUC

0.81 → 0.83

+2%



TEAM CONTRIBUTIONS

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1

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Individual Contributions:

- Feature validation
- Worked on slides
- Data pipeline implementation

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